

Relation Algebra

§12. Combinatorial functions

(12a)

definition	type	note
$[A] ::= \text{nil} \mid \text{cons}(A, [A])$	$\mathcal{A} \rightarrow \mathcal{A}$	The list type, under the short name it keeps.
$\text{list}(R) = ([\text{nil}, (R \times 1) \text{cons}])$	$[A] \rightarrow [B]$	The relator's action on $R : A \rightarrow B$: one R per element, the shape untouched.
$\text{subseq} = ([\text{nil}, \text{cons} \circ \pi_2])$	$[A] \rightarrow [A]$	$xs \text{ subseq } ys$: ys is xs with elements dropped — cons keeps the head, π_2 drops it.
$\text{prefix} = ([\text{nil}, \text{nil} \circ \text{cons}])$ $\text{prefix} = \text{cat} \circ \pi_1 = \text{init}^*$	$[A] \rightarrow [A]$	ys is an initial segment of xs ; the first nil is where it stops early. $\text{init} \triangleq \text{snoc} \circ \pi_1$.
$\text{suffix} = \text{cat} \circ \pi_2 = \text{tail}^*$	$[A] \rightarrow [A]$	The dual, $\text{tail} \triangleq \text{cons} \circ \pi_2$; as a reduce it needs snoc -lists.
$\text{segment} = \text{suffix} \text{ prefix}$	$[A] \rightarrow [A]$	A contiguous stretch of xs : a suffix, then a prefix of that.
$\text{partition} = \text{concat}^*$	$[A] \rightarrow [[A]]$	This cat is restricted to $[A]^* \times [A] \rightarrow [A]$, so ys is a list of non-empty segments of xs .
$\text{concat} = ([\text{nil}, \text{cat}])$	$[[A]] \rightarrow [A]$	Joins the segments back up, which is why its converse splits a list.
inits	$[A] \rightarrow [[A]]$	Implements $\frac{\text{prefix}}{\ni}$, listing the prefixes by increasing length.
tails	$[A] \rightarrow [[A]]$	Implements $\frac{\text{suffix}}{\ni}$ by decreasing length — the opposite order.
$\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\ni} \text{est}(R^\circ)$	$[A] \rightarrow [A]$	The longest subsequence of xs whose every element passes p . $\text{est}(R^\circ)$ is $\triangleq \text{est}$
$R \triangleq \text{length} \leq \text{length}^\circ$	$[A] \rightarrow [A]$	The preorder filter and takewhile maximise over: the longer list wins. $\geq \triangleq \leq^\circ$
$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\ni} \text{est}(R^\circ)$	$[A] \rightarrow [A]$	The same with prefix for subseq : the longest prefix whose every element passes p .
$\text{mss} \triangleq \frac{\text{segment sum}}{\ni} \text{est}(\geq)$	$[A] \rightarrow A$	Maximum segment sum. $\text{segment} = \text{suffix prefix}$ splits it into $\frac{\text{prefix sum}}{\ni} \text{est}(\geq)$ on each suffix.

§12.1. $\frac{\text{subseq}}{\ni} = ([\text{nil} \frac{1}{\ni}, (\frac{1 \times \ni}{\ni} \text{E}(\text{cons}), \pi_2) \text{cup}])$

(12.1a)

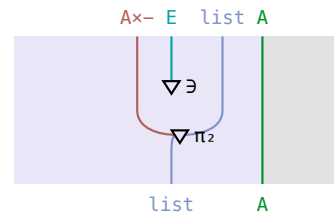
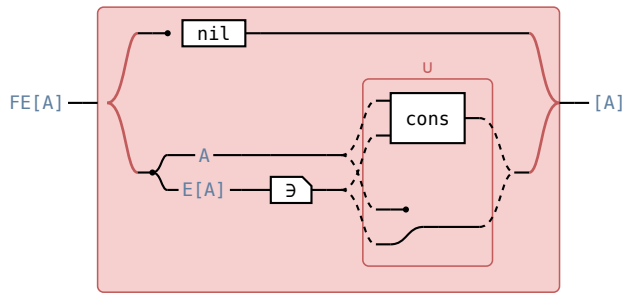
$\text{cup} \triangleq \frac{\pi_1 \ni \cup \pi_2 \ni}{\ni} : EA \times EA \rightarrow EA$, so $\frac{R \cup S}{\ni} = (\frac{R}{\ni}, \frac{S}{\ni}) \text{cup}$.

$$\frac{F(\exists)[\text{nil}, \text{cons } u \ \pi_2]}{\exists} = [\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \ \pi_2)}{\exists}] \quad (12.1b)$$

the set of lists the algebra builds is, from nothing, just **nil**, and from a head and a set of tails, every tail in the set with the head put on or left off — cata-map-calc at **subseq** = $([\text{nil}, \text{cons } u \ \pi_2])^\exists$, $\triangleq$ **subseq**.

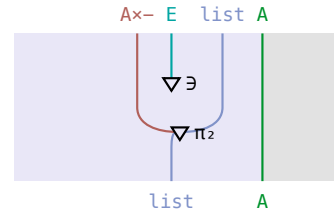
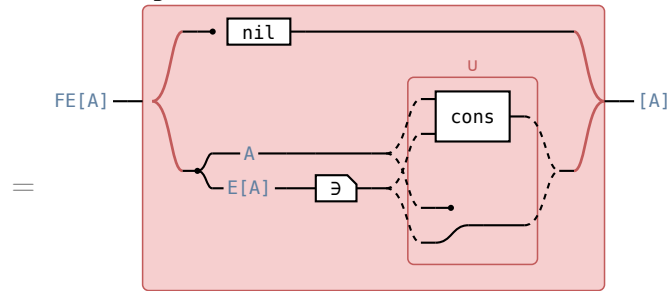
circuit — the fork is $F([A]) = 1 + A \times [A]$: **nil** above, the pair below

Hinze–Marsden



$$\frac{F(\exists)[\text{nil}, \text{cons } u \ \pi_2]}{\exists}$$

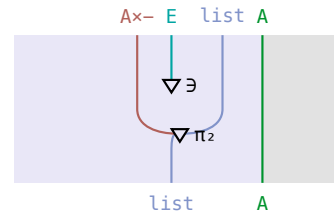
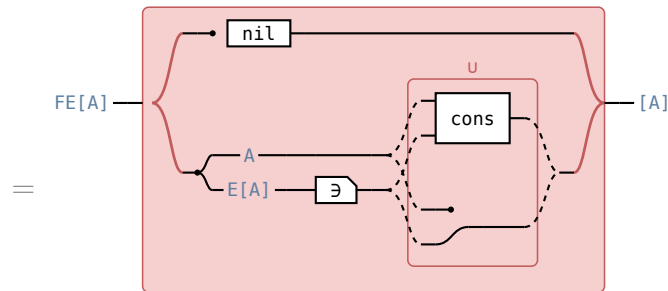
the π_2 operand of $\text{cons } u \ \pi_2$ under the $1 \times \exists$ summand of $F(\exists)$, i.e. $(1 \times \exists) \pi_2$



$$\frac{(1+1 \times \exists)[\text{nil}, \text{cons } u \ \pi_2]}{\exists}$$

$$F(X) = 1 + A \times X \triangleq \text{subseq}$$

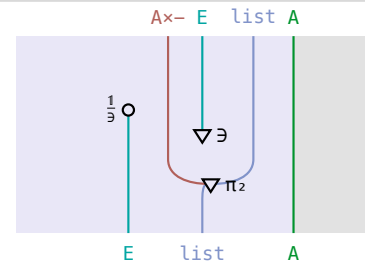
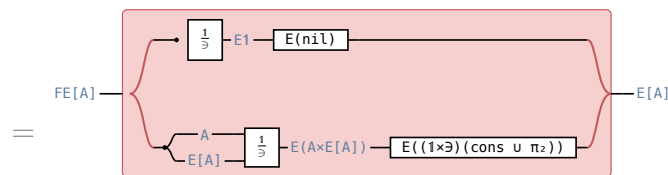
the same operand under $1 + 1 \times \exists$, whose $1 \times \exists$ summand it sits in



$$\frac{[\text{nil}, (1 \times \exists)(\text{cons } u \ \pi_2)]}{\exists}$$

$$R + S \triangleq [Rl, Sr], \quad l[R, S] = R, \quad r[R, S] = S \quad \text{— coprod-laws}$$

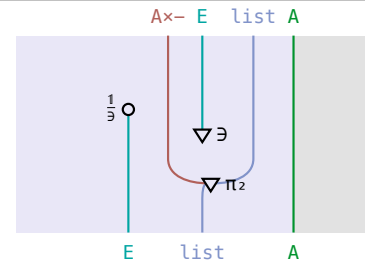
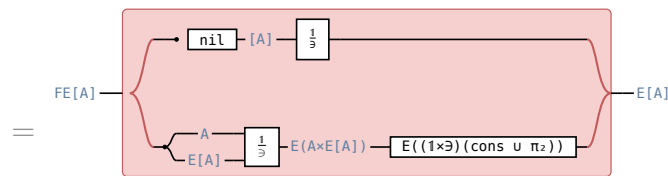
the π_2 operand of the second arm $(1 \times \exists)(\text{cons } u \ \pi_2)$



$$[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \ \pi_2)}{\exists}]$$

$$\text{coprod-calc at } T := [\text{nil}, (1 \times \exists)(\text{cons } u \ \pi_2)]$$

the π_2 operand under its $1 \times \exists$, the arm (12.1c)'s square rewrites, $(1 \times \exists) \pi_2 = \pi_2 \exists$



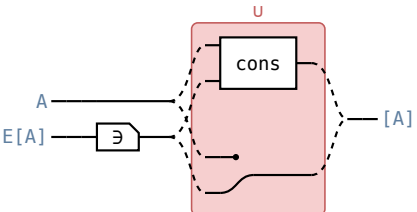
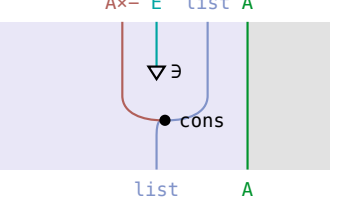
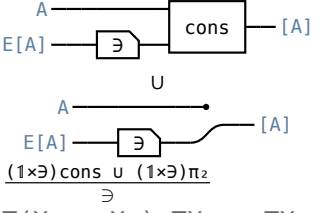
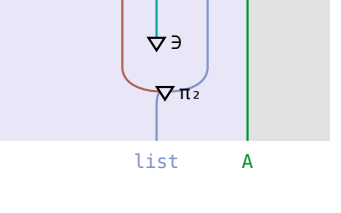
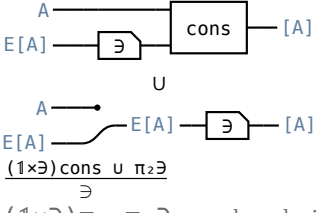
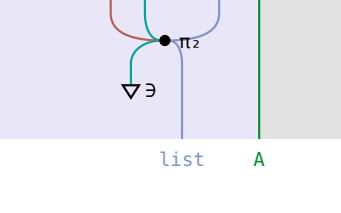
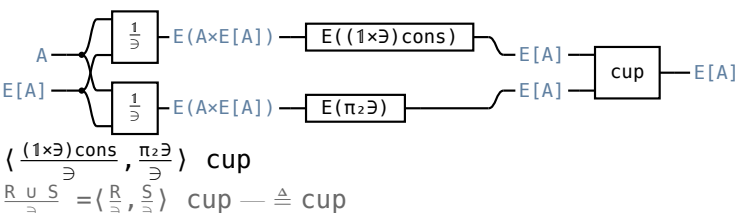
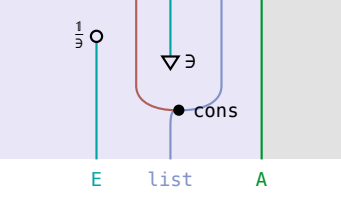
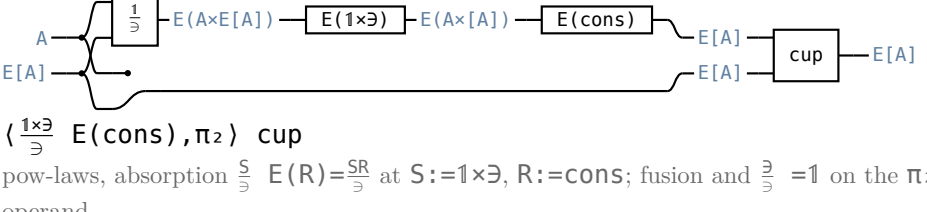
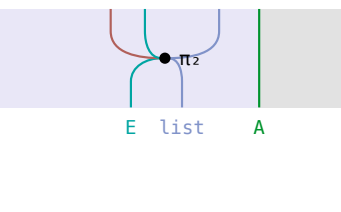
$$[\text{nil } \frac{1}{\exists}, \frac{(1 \times \exists)(\text{cons } u \ \pi_2)}{\exists}]$$

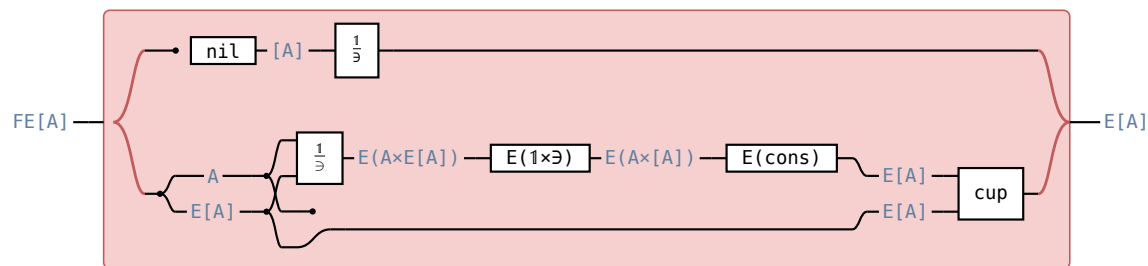
$$\text{pow-laws, } \frac{f}{\exists} = f \frac{1}{\exists} \text{ for } f \text{ a map, at } f := \text{nil}$$

the same operand; the two rows differ only in the **nil** arm

$$\begin{array}{ccc} A \times E[A] & \xrightarrow{\pi_2} & E[A] \\ \downarrow 1 \times \exists & & \downarrow \exists \\ A \times [A] & \xrightarrow{\pi_2} & [A] \end{array}$$

(12.1c)

$\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists} = \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ power transpose of join: the power transpose of the join of two relations is $\left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}$, where cup is the function that returns the union of two sets	(12.1d)
circuit	Hinze–Marsden
 $\frac{(1 \times \exists)(\text{cons} \cup \pi_2)}{\exists}$ (12.1b)'s second branch	 the cons operand of cons $\cup$ π_2
 $\frac{(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2}{\exists}$ $T(X_1 \cup X_2) = TX_1 \cup TX_2$ — adj-cross	 the π_2 operand of $(1 \times \exists)\text{cons} \cup (1 \times \exists)\pi_2$
 $\frac{(1 \times \exists)\text{cons} \cup \pi_2 \exists}{\exists}$ $(1 \times \exists)\pi_2 = \pi_2 \exists$ — relprod-pic at π_2, an equality because 1 is entire	 the π_2 operand of $(1 \times \exists)\text{cons} \cup \pi_2 \exists$
 $\frac{(1 \times \exists)\text{cons}, \pi_2 \exists}{\exists} \text{cup}$ $\frac{R \cup S}{\exists} = \left(\frac{R}{\exists}, \frac{S}{\exists} \right) \text{cup}$ — $\triangle$ cup	 the cons operand under its 1%
 $\left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup}$ pow-laws, absorption $\frac{S}{\exists} = \frac{SR}{\exists}$ at $S := 1 \times \exists$, $R := \text{cons}$; fusion and $\frac{\exists}{\exists} = 1$ on the π_2 operand	 the π_2 operand, bare π_2



$$\left[\frac{\text{nil}}{\exists}, \left(\frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right) \text{cup} \right]$$

which writes **Pcons**; **cons** is a map, and there $P(\text{cons}) = E(\text{cons})$ — powrel-laws.